

ANDREW PHENG QI JINN

Software Engineering graduate (CGPA 3.93) seeking software engineering or AI-related roles, with hands-on experience building web and mobile apps, AI-integrated systems, and an award-winning wearable assistive device for visually impaired users.

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EDUCATION

Bachelor of Software Engineering (Honours), Tunku Abdul Rahman University of Management and Technology, Penang, Malaysia.

Jul 2024 - Jul 2026

CGPA of 3.9304, President's List, MUET Band 5.0, 100% Scholarship

- Assignment Projects: E-Commerce Website for Healthcare Products, Automated Software Testing on Agoda Website, Workshop Management Mobile Application for Car Mechanics
- Courses include Software Engineering, Data Structures and Algorithms, Web-Based Integrated Systems, Advanced Database Management, Mobile Application Development, Software Testing, Software Project Management, Software Design and Architecture, Image Processing, Software Maintenance, Agile Software Development, Artificial Intelligence
- Final Year Project: VISION (see **Projects**)

Diploma in Computer Science, Tunku Abdul Rahman University of Management and Technology, Penang, Malaysia.

Jul 2022 - Jun 2024

CGPA of 3.99, President's List, 50% TAR UMT Merit Scholarship, 50% Hong Leong Foundation Scholarship

- Assignment Projects: Multilevel Marketing System, Networking Configurations (CISCO), E-commerce Website for Health and Wellness Products
 - Courses include Web Design and Development, System Analysis and Design, Database Development and Applications, Object-Oriented Programming Techniques, GUI
 - Professional Certifications: Cisco Networking Academy
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WORK EXPERIENCE

R&D Intern

Guangdong Gospower Electric Technology Co., Ltd., Dongguan, China, Jan 2026 - Jul 2026

- Performed PMBus communication and telemetry verification, including software monitoring, register validation, and data comparison against hardware measurements.
- Conducted power supply testing and validation using oscilloscopes, electronic loads, and power analysers to analyse electrical performance.
- Assisted in hardware debugging and reliability testing (HALT, temperature/humidity, noise analysis).

Software Developer (Intern)

SiP Technology (M) Sdn Bhd, Penang, Nov 2023 - Jan 2024

- Led a team of 3 to migrate and refine legacy software into a modern WPF application using C#, XAML, and JSON configuration, delivering within a 10-week deadline.
 - Developed a GUI application to visualise and manage PCB panel dispensing processes, featuring multiple display modes, recipe-based event handling, and real-time data logging.
 - Implemented TCP client-server communication for command data exchange, an admin authentication system with role-based access control, and comprehensive input validation.
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KEY PROJECTS

VISION: Visually Impaired Support with Intelligent Object-Recognition and Navigation

Final Year Project, Jun 2025 - Nov 2025

- Developed AI-powered smart glasses with a custom-trained YOLOv8 model, featuring object detection, obstacle alerts (sound and vibration), voice-guided navigation, multi-face recognition, document scanning with AI summaries, and two-way voice messaging, hands-free in English, Mandarin, and Malay.
- Built a companion Flutter mobile app with dual interfaces for visually impaired users and their caretakers, featuring location tracking, location history, rich messaging (text, voice, image, video), familiar face management, favourite locations, document history, secure QR-based pairing, and Firebase cloud sync.
- Tech Stack: Python, Dart/Flutter, YOLOv8, OpenCV, Vosk, OpenAI Vision API, Google Maps API, Firebase (Firestore, Storage, Cloud Messaging), Raspberry Pi 5, sensors
- Won **Best Technology Award at TAR UMT EETPIX 2026** (FYP Exhibition Day) and selected as **finalist** for the **2026 Urban Innovation and Entrepreneurship Competition by GISU** in Guangzhou, China.

Multilingual Cyberbullying Detection System

Artificial Intelligence Coursework, Jul 2025 - Sep 2025

- Trained and evaluated 8 models including CNN, BLSTM with Attention (TensorFlow/Keras), Logistic Regression, Naive Bayes, and XGBoost (scikit-learn) on ~196,000 samples for binary cyberbullying text classification.
- Implemented TF-IDF vectorization, stratified cross-validation, threshold optimization for recall-first strategy, and full evaluation metrics (ROC, precision-recall, confusion matrix, calibration curves).
- Fine-tuned transformer models (BERTweet, BART, T5) for multilingual detection across English, Malay, and Manglish text.

Mental Health and Productivity System: Virtual Forest Module

Collaborative Development Coursework, Feb 2025 - May 2025

- Designed and developed the Virtual Forest gamification module for a mental health and productivity web application, featuring three focus session modes, a tiered reward system with four rarity levels of tree boxes, and a pity system guaranteeing rare drops after consecutive failed attempts.
- Implemented anti-exploitation mechanics (withered trees for abandoned sessions), an in-game economy, inventory management, customisable forest plots with biomes, and community features
- Tech Stack: HTML, CSS, JavaScript, Bootstrap, Java.

CO-CURRICULAR INVOLVEMENTS

StartupPenang500@University, May 2025 - Nov 2025

- Pitched VISION at StartupPenang500@University, achieving **1st Runner-Up** at Campus Tech X Challenge 2025 and presenting the MVP at Startup Venture Vibes (SVV) Summit 2025 on 27 November 2025.

Jabil Early Career Programs (ECP) Bootcamp 2.0, June 2025

- Received professional mentorship and networking opportunities with industry subject matter experts and leaders, as well as experienced hands-on projects applying Data Science and SAP concepts to address real-world business challenges.
- Delivered a case study presentation to industry subject matter experts and won **Second-Runner Up**.

SKILLS AND COMPETENCIES

Technical Skills

- **Programming Languages:** Java, Python, C, C#, Dart, Assembly Language
- **Frameworks & Libraries:** Flutter, YOLOv8, OpenCV, TensorFlow/Keras, scikit-learn, PyTorch, Hugging Face Transformers
- **Web Development:** HTML, CSS, PHP, JavaScript, JSP
- **Databases & Cloud:** Oracle SQL, PL/SQL, Firebase Firestore, Firebase Storage, SQLite
- **Tools & Platforms:** Linux, Git, Automated Software Testing, Network Configuration (Cisco), Microsoft Project, Jira

Language Competencies

- Competent and fluent in English, Bahasa Malaysia and Mandarin

Soft Skills

- Motivated, Communication, Leadership, Teamwork, Critical Thinking, Adaptability, Creativity, Presentation

REFERENCES

Ms. Kee Sim Ee

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